



Evaluation of (vapor + liquid) equilibria for the binary systems (1-octanol + cyclohexane) and (1-octanol + *n*-hexane), at low alcohol compositions

Gabriel Ovejero*, M. Dolores Romero, Eduardo Díez, Tânia Lopes, Ismael Díaz

Grupo de Catálisis y Procesos de Separación (CyPS), Departamento de Ingeniería Química, Facultad de C. Químicas, Universidad Complutense de Madrid, Avda. Complutense s/n, 28040 Madrid, Spain

ARTICLE INFO

Article history:

Received 17 April 2008

Received in revised form 12 June 2008

Accepted 12 June 2008

Available online 20 June 2008

Keywords:

(Vapor + liquid) equilibrium

Cyclohexane

n-Hexane

1-Octanol

ABSTRACT

Isobaric (vapor + liquid) equilibrium at $p = 101.32$ kPa of pressure has been determined for the systems (1-octanol + cyclohexane) and (1-octanol + *n*-hexane), at low alcohol mole fractions. These data were satisfactorily correlated, using ASPEN PLUS® commercial software, with Wilson, NRTL, and UNIQUAC activity coefficient models to obtain the binary interaction parameters of both mixtures. Also, UNIFAC group contribution method was employed to predict the equilibrium of both mixtures. With regression values an accurate knowledge of (vapor + liquid) equilibrium for both mixtures can be reached in a range of 1-octanol mole fractions less than 0.1. UNIFAC method provides acceptable results for (1-octanol + *n*-hexane) system but not for (1-octanol + cyclohexane) system.

© 2008 Elsevier Ltd. All rights reserved.

1. Introduction

In many industrial processes, distillation type operations are extensively used for the separation of a great number of liquid mixtures or the purification of many products. In order to accurately design the necessary equipment, knowledge of (vapor + liquid) equilibrium is very important, to calculate, for example, the number of theoretical stages in a tray column. Moreover, it is of great importance to obtain the equilibrium data in the range of concentrations in which the separation is being developed.

Cyclohexane and *n*-hexane are important organic solvents employed in many processes, and equilibrium data with both compounds were obtained. On the other hand, 1-octanol belongs to one of the most important additives group, called polar modifiers. High and medium molecular weight alcohols have been demonstrated to act properly as polar modifiers, due to their hydroxyl group at the end of a large hydrocarbon chain.

Because the last objective of the project is to develop a simulation model of (vapor + liquid + liquid) distillation process in which additives molar compositions are below 10%, to obtain a more accurate definition, only this range of the equilibrium curve was considered. 1-Octanol was chosen as additive, instead of other alcohols with lower molecular weight, because the distillation pro-

cess also involves the separation between an organic and an aqueous phase and the solubility in water of lower molecular weight alcohols could be not negligible.

Equilibrium of binary mixtures of 1-octanol and other alcohols such as methanol and ethanol [1] was determined previously. Furthermore, ternary mixtures of (1-octanol + methanol + water) [2] and (1-octanol + ethanol + water) [3] was studied too. In another publication [4], equilibrium of 1-octanol with 1-bromobutane was also obtained. Concerning the mixtures involved in this work, there are two previous published references [5,6], but none of them deal with data the range of low alcohol compositions. The lower values for 1-octanol liquid mole fraction are 0.1, in the first one, and 0.088, in the second one.

This paper reports the results of measurements of (vapor + liquid) equilibrium for the binary systems (1-octanol + cyclohexane) and (1-octanol + *n*-hexane) at low alcohol compositions and at atmospheric pressure. All equilibrium data were fitted to Wilson [7], NRTL [8], and UNIQUAC [9] activity coefficients models, to calculate the binary interaction parameters of these models. Adjustments were performed using ASPEN PLUS® commercial software; this way, when the adjustments are done with the same platform that the future distillation model, problems of different units or different equation for the models could be avoided.

The results of the adjustments are compared with UNIFAC group contribution method [10]. Predictions methods are frequently useful for initial approximations, so in most of the cases is necessary to obtain accurate experimental results.

* Corresponding author. Tel.: +34 91 394 4111; fax: +34 91 394 4114.

E-mail address: govejero@quim.ucm.es (G. Ovejero).

2. Experimental

2.1. Materials

Cyclohexane, *n*-hexane, and 1-octanol were purchased from Aldrich with nominal mass fraction purities >0.997, >0.99, and >0.990 respectively, and were used without any purification step.

2.2. Apparatus and procedure

The experiments were carried out in an apparatus, which had been previously used before [11–13]. In this apparatus, entirely made of glass, both the liquid and vapor phases are continuously recirculated in order to provide mixing of the phases and to ensure that equilibrium was established. The vapor condenser is connected to a constant-pressure system controlled by a Cartesian manostat. Pressure was kept constant at $p = 101.32$ kPa, with an accuracy of ± 0.1 kPa. The equilibrium temperatures were measured with a certified thermocouple type J, with an accuracy of $\pm 0.1^\circ\text{C}$. Each equilibrium point was sampled twice to assure the quality of the results.

The liquid and condensed vapor analyses were carried out by gas chromatography on a Perkin–Elmer A/S gas chromatograph equipped with a J&W DB-23 capillary column and a flame ionization detector. Each sample was analyzed twice.

3. Results and discussion

In tables 1 and 2 (vapor + liquid) equilibrium data (x_1, y_1, T) and experimental liquid phase activity coefficients are presented for both mixtures studied in the work. These coefficients were calculated from equations (1) and (2), which uses virial equation, truncated after the second term, for vapor phase fugacity coefficients.

$$\gamma_i = (y_i \cdot p / x_i \cdot p_i^\circ) \cdot \exp [(B_{ij} - V_i) \cdot (p - p_i^\circ) + (1 - y_i) \cdot p \cdot \delta_{ij} / R \cdot T] \quad (1)$$

$$\delta_{ij} = 2B_{ij} - B_{ii} - B_{jj}, \quad (2)$$

where γ_i is the activity coefficient, x_1 and y_1 are the liquid and vapor equilibrium mole fraction, p is the total pressure, p_i° is the pure component vapor pressure, V_i is the liquid molar volume, R is the universal gas constant, T is the absolute temperature, B_{ii} is the second virial coefficient of the pure gas, and B_{ij} is the cross second virial coefficient. Virial coefficients were calculated according to Tsonopoulos correlation [14] and vapor pressures of the pure components were calculated with Antoine equation (equation (3)), where Antoine constants for the three compounds, taken from Reid and Prausnitz [15], are shown in table 3. Also, molar volumes of the pure compounds were taken from Reid and Prausnitz [15].

$$\ln p_i^\circ / \text{kPa} = A + B / [T / \text{K} + C]. \quad (3)$$

Table 1

(Vapor + liquid) equilibrium data for {1-octanol (1) + cyclohexane (2)} at $p = 101.3$ kPa

x_1	y_1	T_b/K	γ_1	γ_2
0.01001	0.00054	354.1	5.72	1.00
0.01330	0.00068	354.1	5.47	1.01
0.01952	0.00082	354.2	4.44	1.01
0.02877	0.00105	354.4	3.79	1.01
0.03519	0.00119	354.5	3.50	1.02
0.03649	0.00120	354.6	3.40	1.02
0.04403	0.00127	354.7	2.96	1.02
0.05645	0.00142	354.8	2.56	1.03
0.05757	0.00146	354.9	2.58	1.03

Table 2

(Vapor + liquid) equilibrium data for {1-octanol (1) + *n*-hexane (2)} at $p = 101.3$ kPa

x_1	y_1	T_b/K	γ_1	γ_2
0.01520	0.00049	342.2	7.34	1.01
0.01588	0.00049	342.2	7.05	1.01
0.01692	0.00052	342.2	6.90	1.01
0.02332	0.00057	342.4	5.45	1.01
0.02401	0.00058	342.4	5.45	1.01
0.02467	0.00058	342.3	5.27	1.01
0.02718	0.00063	342.4	5.18	1.01
0.03668	0.00070	342.4	4.27	1.02
0.03951	0.00068	342.4	3.84	1.03
0.04052	0.00069	342.4	3.79	1.03
0.04917	0.00073	342.5	3.31	1.03
0.05192	0.00073	342.5	3.13	1.04
0.05457	0.00082	342.6	3.33	1.04
0.06575	0.00091	342.6	3.06	1.05

Table 3

Antoine equation parameters of the pure substances

Compound	A	B	C	$T_{\min}/\text{K}$	$T_{\max}/\text{K}$
Cyclohexane	13.74	−2766.63	−50.5	280	380
<i>n</i> -Hexane	13.82	−2697.55	−48.78	245	370
1-Octanol	13.73	−3017.81	−137.1	343	468

As it was expected, the resulting vapor phase fugacity coefficients in both mixtures were near unity, because all the data belong to the dilute area of the equilibrium curve. As it can be seen in tables 1 and 2, both systems show positive deviation from ideality.

The experimental data were tested to evaluate the thermodynamic consistency by using the method described by Fredenslund et al. [16]. To develop the test, a four-term Legendre polynomial was used to correlate the excess Gibbs free energy. According to this criterion, the experimental data are consistent if the mean absolute deviation between calculated and measured mole fractions of component 1 in the vapor phase, Δy , is less than 0.01. In this case, the values obtained are $\Delta y = 0.0047$ for (1-octanol + cyclohexane) system and $\Delta y = 0.0059$ for (1-octanol + *n*-hexane) system.

The consistency was also confirmed by using Wisniak and Tamir's modification [17] of the McDermott–Ellis test [18]. According to this test, thermodynamic consistency is confirmed if $D < D_{\max}$. For (1-octanol + cyclohexane), values of $D = 0.12$ and $D_{\max} = 0.13$ were obtained. For (1-octanol + *n*-hexane), values of $D = 0.16$ and $D_{\max} = 0.19$ were determined.

The results of both tests indicate that the all data obtained are thermodynamically consistent.

Although there are some experimental data of (1-octanol + *n*-hexane) already published [6], it is difficult to establish a comparison because in this case the equilibrium was determined isothermally at $T = 313.15$ K. At this temperature, the authors obtained a value of 3.99 for the 1-octanol activity coefficient, when 1-octanol liquid mole fraction of 0.0880 and total pressure is $p = 35.923$ kPa. Despite of the different temperatures, this activity coefficient value is close to the one in this paper (3.06) for the lowest 1-octanol mole fraction tested.

The equilibrium data were correlated with the Wilson [7], NRTL [8], and UNIQUAC [9] activity coefficients models using ASPEN PLUS® commercial software. To develop the calculations, an objective function called “maximum likelihood” was used. This function minimizes the difference between the experimental and fitted data for all the variables that appear in adjustment (pressure, temperature, and liquid and vapor composition of both components).

The expression for activity coefficients of Wilson model that this software utilizes to calculate the activity coefficient in a multicomponent mixture is described in equation (4)

$$\ln \gamma_{ij} = 1 - \ln \left(\sum_j A_{ij} \cdot x_j \right) - \sum_j A_{ji} \cdot x_j / \sum_k A_{jk} \cdot x_k. \quad (4)$$

In this expression, x_i represents liquid mole fraction of component i , and the binary interaction parameters (A_{ij}, A_{ji}) are defined as a function of temperature in equation (5)

$$\ln A_{ij} = \ln(V_j/V_i) + b_{ij}/T, \quad (5)$$

where V_i and V_j are the liquid molar volumes at the system temperature calculated using the Rackett [19] model.

For NRTL model, the expression for calculating the activity coefficients is described in equation (6)

$$\ln \gamma_{ij} = \left(\sum_{j=1}^C \tau_{ji} G_{ji} x_j / \sum_{k=1}^C x_k G_{ki} \right) + \sum_{j=1}^C \left(x_j G_{ij} / \sum_{k=1}^C x_k G_{kj} \right) \left[\tau_{ij} - \left(\sum_{n=1}^C \tau_{nj} G_{nj} x_n / \sum_{k=1}^C x_k G_{kj} \right) \right]. \quad (6)$$

As before, x_i represents liquid mole fraction of component i , and the binary interaction parameters (τ_{ij}, τ_{ji}) are defined as a function of temperature in equation (7). The “non randomness factor”, α , values [8] range from 0.2 to 0.47.

$$G_{ij} = \exp(-\alpha_{ij} \tau_{ij}), \quad \tau_{ij} = a_{ij} + b_{ij}/T, \quad \alpha_{ij} = c_{ij}. \quad (7)$$

At last, for UNIQUAC model, the values of the activity coefficients are calculated according to:

$$\ln \gamma_{ij} = \ln \Phi_i/x_i + z/2 \cdot q_i \ln(\theta_i/\Phi_i) - q'_i \ln t'_i - q'_i \sum_j \theta'_j \cdot \tau_{ij}/t' + l_i + q'_i - (\Phi_i/x_i) \cdot \sum_j x_j l_j. \quad (8)$$

In this case, the binary interaction parameters are defined as a function of temperature according to

$$\tau_{ij} = \exp(a_{ij} + b_{ij}/T + c_{ij} \cdot \ln(T) + d_{ij} \cdot T) \quad (9)$$

and the other parameters are structural ones, defined for each component [9].

In tables 4 and 5 Wilson, NRTL, and UNIQUAC b_{12} and b_{21} adjusted constants of the binary interaction parameters, along with the standard deviations (to ensure the quality of the values), are shown.

As example, figure 1 shows the experimental activity coefficients along with those regressed using for Wilson model; similar plots can be obtained for NRTL and UNIQUAC models.

In order to compare experimental and regressed data for Wilson model, both results are compared in figure 2. As it can be seen, Wilson model accurately fit the experimental data for both mixtures by considering only the b_{ij} values of the binary interaction parameters; for this reason it is not necessary to introduce more adjusting values. With UNIQUAC and NRTL models, the results of the regression are also adequate.

To support these statements and to determine the quality of the adjustments, in table 6, the mean percentage deviations of temperature ($\sigma(T) = [T(\text{expt.}) - T(\text{calc.})]^2$), 1-octanol vapor mole fraction

Table 4
 b_{12} and b_{21} coefficients of the binary interaction parameters of Wilson, NRTL, and UNIQUAC models for {1-octanol (1) + cyclohexane (2)}

Model	b_{12}	σ_{12}	b_{21}	σ_{21}
Wilson	−1161.4	33.7	248.9	7.7
NRTL	−303.1	11.2	1204.6	40.2
UNIQUAC	265.7	7.5	−545.3	19.1

σ_{ij} , standard deviation of b_{ij} .

Table 5

b_{12} and b_{21} coefficients of the binary interaction parameters of Wilson, NRTL, and UNIQUAC models for {1-octanol (1) + *n*-hexane (2)}

Model	b_{12}	σ_{12}	b_{21}	σ_{21}
Wilson	−1213.6	36.0	203.1	5.7
NRTL	561.6	13.4	1686.9	46.2
UNIQUAC	266.6	5.8	−567.2	15.8

σ_{ij} , standard deviation of b_{ij} .

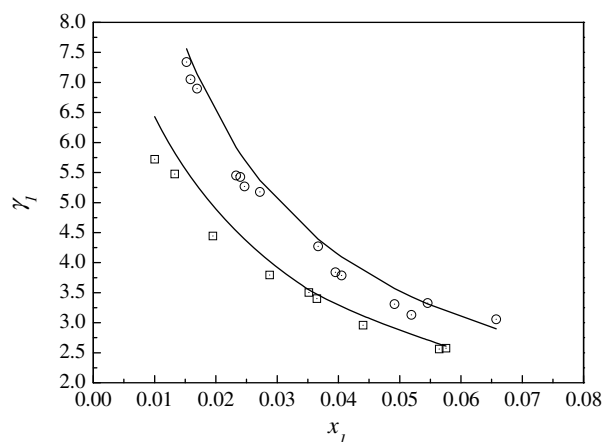


FIGURE 1. 1-Octanol activity coefficients γ_1 as a function of 1-octanol mole fraction x_1 ; (□) experimentally determined for (1-octanol + cyclohexane) system $p = 101.32$ kPa; (○) experimentally determined for (1-octanol + *n*-hexane) system at $p = 101.32$ kPa; (—) values obtained with Wilson model.

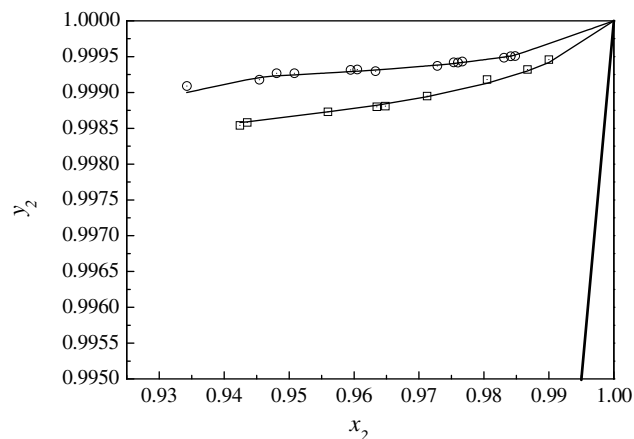


FIGURE 2. Vapor mole fractions y_2 as a function of cyclohexane liquid mole fractions x_2 at $p = 101.32$ kPa; (□) liquid cyclohexane mole fraction as a function of vapor cyclohexane mole fraction for (1-octanol + cyclohexane) system; (○) liquid *n*-hexane mole fraction as a function of vapor *n*-hexane mole fraction for (1-octanol + *n*-hexane) system; (—) values regressed with Wilson model.

($\sigma(y_1) = [y_1(\text{expt.}) - y_1(\text{calc.})]^2$) and 1-octanol activity coefficients ($\sigma(\gamma_1) = [\gamma_1(\text{expt.}) - \gamma_1(\text{calc.})]^2$) are presented.

With the obtained parameters, an accurate knowledge of (vapor + liquid) equilibrium for both mixtures is achieved in the range of less than 0.01 mole fractions of 1-octanol.

Figure 3 shows a comparison between experimental data and those predicted by UNIFAC group contribution method [10]. It is shown that only in the case of (1-octanol + *n*-hexane) system, the experimental data are satisfactorily predicted by the UNIFAC method.

Table 6

Temperature and root-mean-square deviations for the methods tested

Method	{1-Octanol (1) + cyclohexane (2)}			{1-Octanol (1) + <i>n</i> -hexane (2)}		
	$100 \cdot \sigma(T)$	$100 \cdot \sigma(y_1)$	$100 \cdot \sigma(\gamma_1)$	$100 \cdot \sigma(T)$	$100 \cdot \sigma(y_1)$	$100 \cdot \sigma(\gamma_1)$
Wilson	0.10	3.54	3.44	0.10	3.17	3.79
NRTL	0.11	4.68	4.97	0.20	4.91	5.57
UNIQUAC	0.11	4.40	4.55	0.18	4.36	5.04

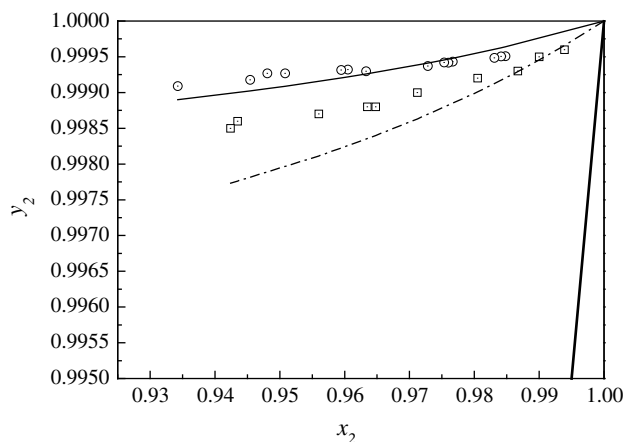


FIGURE 3. Vapor mole fractions y_2 as a function of liquid mole fractions x_2 at $p = 101.32$ kPa; (□) liquid cyclohexane mole fraction as a function of vapor cyclohexane mole fraction for (1-octanol + cyclohexane) system; (○) liquid *n*-hexane mole fraction as a function of vapor *n*-hexane mole fraction for (1-octanol + *n*-hexane) system; (---) values predicted with UNIFAC model for (1-octanol + cyclohexane) system; (—) values predicted with UNIFAC model for (1-octanol + *n*-hexane) system.

4. Conclusion

In order to obtain an accurate prediction of the alcohol dilute zone of the equilibrium curve, experimental VLE data were determined at $p = 101.3$ kPa for two binary systems: (1-octanol + cyclohexane) and (1-octanol + *n*-hexane) at low alcohol mole fractions. Wilson, NRTL, and UNIQUAC activity coefficient models accurately fit the experimental data, which allow using the experimental binary interaction parameters obtained, to develop (vapor + liquid)

equilibrium calculations of both mixtures, in the studied composition range. UNIFAC group contribution method provides acceptable results for (1-octanol + *n*-hexane) system, but that is not the case of (1-octanol + cyclohexane) system.

Acknowledgement

Thermodynamic and simulation calculations were performed using the programme ASPEN PLUS®.

References

- [1] A. Arce, A. Blanco, A. Soto, J. Tojo, J. Chem. Eng. Data 40 (1995) 1011–1014.
- [2] A. Arce, J.C. Pérez, A. Soto, Fluid Phase Equilibr. 129 (1997) 187–195.
- [3] A. Arce, J. Martínez-Ageitos, A. Soto, Fluid Phase Equilibr. 122 (1996) 117–129.
- [4] R. Garriga, S. Martínez, P. Pérez, M. Gracia, J. Chem. Eng. Data 49 (2004) 224–230.
- [5] D.C. Minh, M. Rue, J. Chem. Eng. Data 18 (1973) 41–44.
- [6] A. Heintz, E. Dolch, R.N. Lichtenthaler, Fluid Phase Equilibr. 27 (1986) 61–79.
- [7] G.H. Wilson, J. Am. Chem. Soc. 86 (1964) 127–130.
- [8] H. Renon, J.M. Prausnitz, AIChE J. 14 (1968) 135–144.
- [9] D.S. Abrams, J.M. Prausnitz, AIChE J. 21 (1975) 116–128.
- [10] A. Fredenslund, R.L. Jones, J.M. Prausnitz, AIChE J. 21 (1975) 1086–1099.
- [11] G.OvejeroA. de Lucas, J.M. Moreno, J. Chem. Eng. Data 37 (1992) 293–295.
- [12] J. Aguado, M.D. Romero, A. Rodríguez, J.M. Ferrer, L.M. Marugán, J. Chem. Eng. Data 41 (1996) 440–442.
- [13] J. Aguado, M.D. Romero, A. Rodríguez, J.M. Ferrer, L.M. Marugán, J. Chem. Eng. Data 41 (1996) 1375–1378.
- [14] C. Tsonopoulos, AIChE J. 20 (1974) 263–272.
- [15] R.C. Reid, J.M. Prausnitz, B.E. Poling, The Properties of Gases and Liquids, fourth ed., McGraw Hill, New York, 1987.
- [16] A. Fredenslund, J. Gmehling, P. Rasmussen, Vapor-liquid Equilibria using UNIFAC. A Group Contribution Method, Elsevier, Amsterdam, 1977.
- [17] J. Wisniak, A. Tamir, J. Chem. Eng. Data 22 (1977) 253–260.
- [18] C. McDermott, S.R.M. Ellis, Chem. Eng. Sci. 20 (1965) 293–296.
- [19] H.G. Rackett, J. Chem. Eng. Data 15 (1970) 514–517.

JCT 08-145